Kestrel SOC Lab Investigation Report

Endpoint authentication and network monitoring validation

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Executive summary

The authorised VirtualBox lab demonstrated collection and search of Windows event logs, Ubuntu host logs, Suricata network events and Splunk platform logs. The evidence supports a working Windows failed logon detection and a completed investigation of a failed SSH password followed by a successful SSH login.

Kestrel Logistics Group is the logistics and supply chain case-study scenario for this portfolio project. These results describe a home lab, not an assessment of a production organisation. No malicious compromise was established.

Validation area	Evidence and conclusion
Windows detection	Six failed logons for NewUser from 127.0.0.1 within ten minutes satisfied the five-failure threshold. Scheduled alert entries were subsequently visible.
Ubuntu collection	Both auth.log and syslog appeared in Splunk after their monitor inputs and read access were configured.
Suricata collection	Three controlled ICMP alerts and two DHCP hostname alerts were displayed. Network telemetry was searchable.
SSH investigation	Ubuntu recorded one failed and one accepted password. Two matching Suricata SSH connections corroborated the network activity.
Project completion	Core evidence is assembled. Production readiness remains outside the demonstrated scope.

Purpose and audience

This report provides a reviewer with the final source inventory, test methods, observed results, detection logic and practical limitations. It separates recorded facts from interpretation and proposed improvements. Evidence references E01 to E10 identify the supplied screenshots or command output.

Time convention

All events discussed occurred on 16 October 2026 unless stated otherwise. Raw Ubuntu and Suricata timestamps use UTC. The supplied Splunk views display British Summer Time, which is UTC plus one hour. Search result counts belong to their individual time windows and must not be added across screenshots.

Environment and final log source inventory

The three VMs share the 192.168.167.0/24 lab network. Kali combines the Splunk receiver and controlled traffic generator. Ubuntu hosts the network sensor and a Universal Forwarder. Windows provides endpoint authentication and service events.

VM	Lab address	Role
Kali	192.168.167.4	Splunk Enterprise receiver; controlled SSH and ICMP test client
Ubuntu	192.168.167.5	Suricata sensor on enp0s8; Linux host logs; Universal Forwarder
Windows11	192.168.167.6	Windows Security and System logs; Universal Forwarder

ID	Origin and source	Splunk index and sourcetype	Validation
LOG-01	Windows11 WinEventLog:Security	windows XmlWinEventLog:Security	4624, 4625 and 4672 observed
LOG-02	Windows11 WinEventLog:System	windows XmlWinEventLog:System	Service Control Manager 7040 observed
LOG-03	Ubuntu /var/log/auth.log	linux syslog	165 events in collection snapshot; SSH outcomes later observed
LOG-04	Ubuntu /var/log/syslog	linux syslog	5,614 events in collection snapshot
LOG-05	Ubuntu /var/log/suricata/eve.json	suricata suricata:json	ICMP, DHCP, SSH and flow evidence
LOG-06	Kali Splunk /opt/splunk/var/log/splunk/	_internal Multiple platform sourcetypes	Inventory includes scheduler, metrics and splunkd logs
LOG-07	Windows Universal Forwarder var/log/splunk/	_internal Multiple platform sourcetypes	Forwarder internal paths visible in inventory

Collection path

Both Universal Forwarders send to 192.168.167.4 on TCP port 9997. The Ubuntu forward-server check reported that destination active. Splunk Web is accessed on port 8000. Windows Security and System inputs use XML rendering and current_only = 1. The demonstrated service account is NT SERVICE\SplunkForwarder, with Event Log Readers access.

The platform inventory query returned 63,692 events grouped into 42 index, host, source and sourcetype combinations in its selected 24-hour window. Those combinations are not 42 distinct log sources, and the count is not a count of Kali events alone. [E10]

Collection validation and dashboard

Ubuntu host collection

The initial btool search returned no auth.log or syslog monitor stanzas. The splunkfwd account was added to the adm group; its identity and ability to read both files were then confirmed. Monitor inputs were configured for index linux with sourcetype syslog. Splunk subsequently showed 165 auth.log events and 5,614 syslog events in the selected 24-hour collection snapshot. This establishes collection for that window. [E05]



Figure 1 Ubuntu host log collection results with both monitored files visible E05

SOC dashboard

The Kestrel SOC dashboard combines Windows failed logons, successful logons, authentication activity and Suricata network alerts. One view showed 14 NewUser failures in its own 24-hour window. Successful SYSTEM service logons and desktop-related accounts appeared separately. These values describe the displayed window; they do not replace the six-event alert test count. [E03]



Figure 2 Kestrel SOC dashboard with Windows authentication panels and activity chart E03

Use a shared time range when comparing panels. Empty alert fields on flow or statistics events are expected because those records are not necessarily detection alerts. The Suricata panel should filter event_type = alert when showing signatures.

Windows failed logon detection

The controlled Windows test produced six failed logons for NewUser from loopback address 127.0.0.1 on Windows11 during the 18:48:40 to 18:58:40 BST window. The search grouped by host, account and source IP and returned failures = 6. This met the configured threshold of five within ten minutes. [E01]

Raw event field	Recorded value
Event ID and record	4625; EventRecordID 24120
Time	17:57:09.004771 UTC / 18:57:09.004771 BST
Account and source	NewUser; 127.0.0.1; Windows11
Logon type	2 — interactive
Status and substatus	0xc000006d; 0xc000006a — bad password
Process	C:\Windows\System32\svchost.exe

Alert configuration and correction

The saved search was enabled and scheduled every minute, but its dispatch bounds were rt-10m and rt-0m. Replacing those real-time bounds with -10m and now made the search a historical ten-minute evaluation. The Add to Triggered Alerts action was enabled. The UI then displayed scheduled Medium severity alert entries at 18:58:01, 18:59:01 and 19:00:09 BST. [E02]

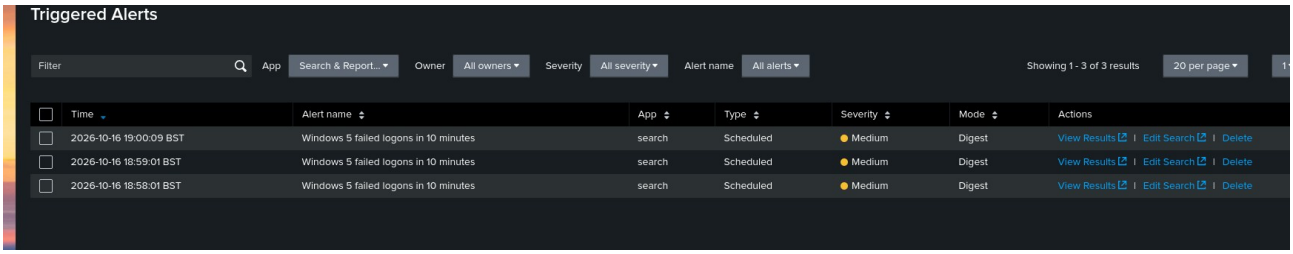


Figure 3 Scheduled Windows detection entries visible in Triggered Alerts E02

These three entries can represent the same activity because the one-minute schedule repeatedly evaluates an overlapping ten-minute window. They are not evidence of three separate incidents. Throttling by host, user and source IP is a proposed improvement after validation.

The loopback source and interactive logon type support a local controlled test. This evidence does not establish a remote attack. Microsoft documents event 4625 as a failed logon record and defines the status fields used here. [R1]

Suricata network validation

Suricata was active and writing eve.json. A status snapshot reported 53,135 rules loaded with no failed rules and 7,245 captured packets with zero kernel drops. These values describe that snapshot, not a guarantee of continuous loss-free monitoring.

Controlled ICMP test

Three echo requests from Kali to Ubuntu generated three LAB ICMP Echo Request alerts. Splunk displayed 18:18:10.651, 18:18:11.649 and 18:18:12.652 BST, source 192.168.167.4 and destination 192.168.167.5. The accompanying ping test received three replies with zero packet loss. The lab rule used signature ID 1000001; the final dashboard displayed severity 3. [E08]

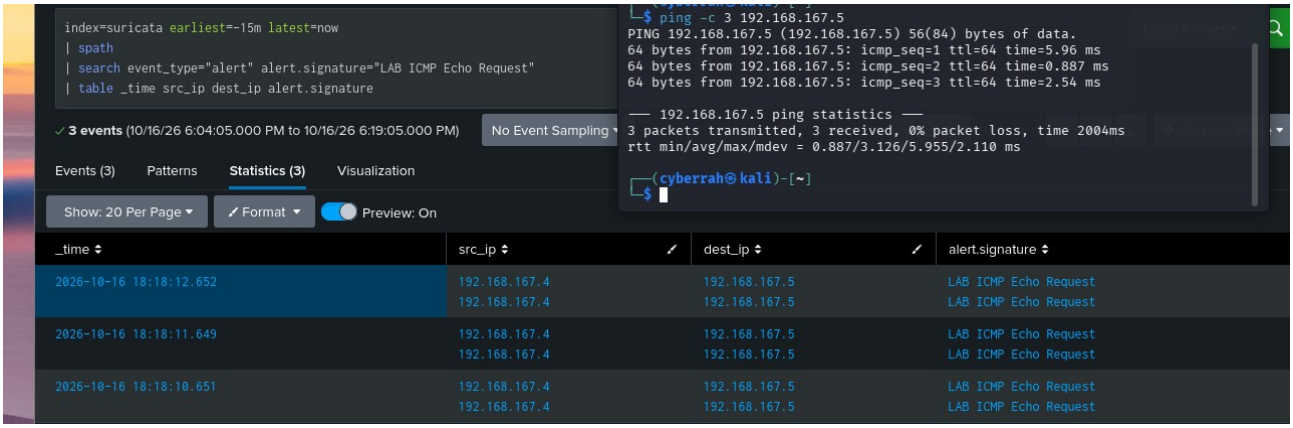


Figure 4 ICMP test traffic and the three corresponding custom Suricata alerts E08

Other alert observations

Two additional events carried the signature ET INFO Possible Kali Linux hostname in DHCP Request Packet. Their recorded source was 0.0.0.0 and destination was 255.255.255.255. The raw alert severity field was 1. The signature describes a hostname observation in DHCP traffic; it does not establish exploitation or compromise. [E09]

Interpretation and coverage

Duplicate values originally shown inside multivalue fields did not represent additional raw events. The later table displayed one source, destination, signature and severity per alert. Any extraction correction should preserve the original raw event.

The tests validate traffic addressed to Ubuntu on the monitored lab interface. They do not prove that the sensor can observe every unicast exchange between Kali and Windows. VirtualBox network placement and traffic visibility must be tested before making broader coverage claims.

SSH investigation across host and network logs

A controlled SSH test from Kali to Ubuntu used account cyberrah. The host fingerprint matched the value obtained directly on Ubuntu. One password attempt was rejected and a subsequent attempt succeeded. Ubuntu recorded the authentication outcomes; Suricata recorded two SSH connections and their flows. [E06, E07]

Evidence	Failed connection	Successful connection
Network tuple	192.168.167.4:39472 → 192.168.167.5:22	192.168.167.4:60232 → 192.168.167.5:22
Ubuntu outcome UTC	Failed password 20:42:47.833179	Accepted password 20:42:56.505503
Suricata flow ID	275744592746539	1421044264735560
Flow start UTC	20:41:28.391881	20:42:53.003182
Flow end UTC	20:42:48.020099	20:50:05.806617
Suricata SSH event BST	21:41:28.686	21:42:53.054
Flow record time BST	21:43:53.195	21:51:14.871

Correlation method

Match the client and server IPs, destination port 22 and the distinct client source ports. Convert the Ubuntu UTC timestamps to the Splunk display timezone, then check that each authentication outcome falls within its corresponding flow start and end interval. Both outcomes meet that check. The accepted password occurred approximately 8.67 seconds after the failed password.

The later flow record timestamps reflect when Suricata reported those flow records. Using only a narrow window around the authentication message could therefore miss the flow records. A search covering 20:40 to 20:53 UTC includes both connections and their later records.

Finding and limits

The evidence contains four relevant Suricata records: two SSH events and two flow events. It supports two connections, not four login attempts. The host log establishes failed and accepted passwords. SSH network metadata cannot establish the password outcome inside encrypted traffic. [R3]

This is authorised validation, not evidence of brute force or account compromise. The earlier 20:31:30 UTC timeout occurred before authentication and is separate from these two password outcomes. The Windows loopback test is also separate; no attack chain between the tests was demonstrated.

SSH evidence and final assessment

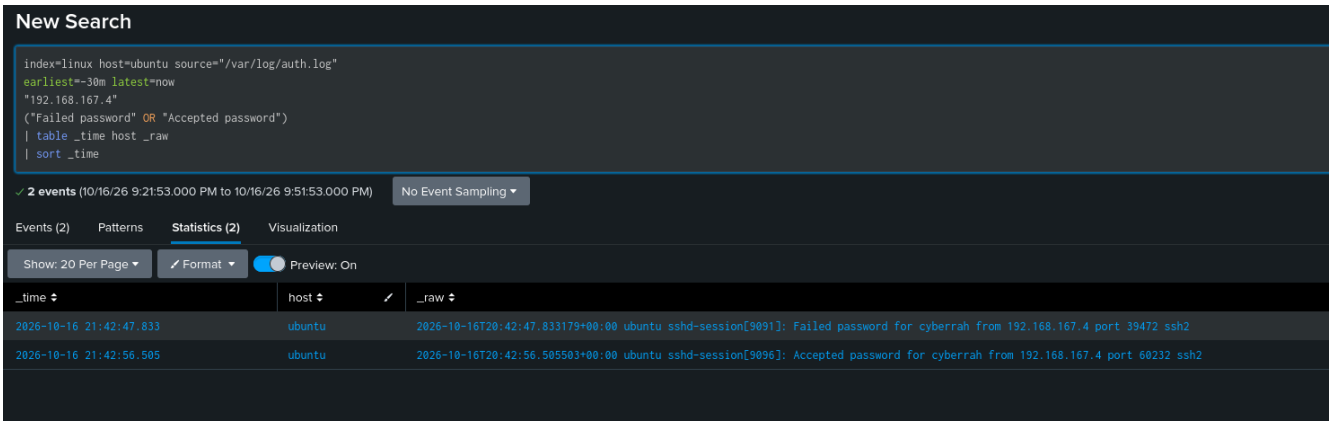


Figure 5 Ubuntu authentication evidence for the failed and accepted SSH password attempts E06

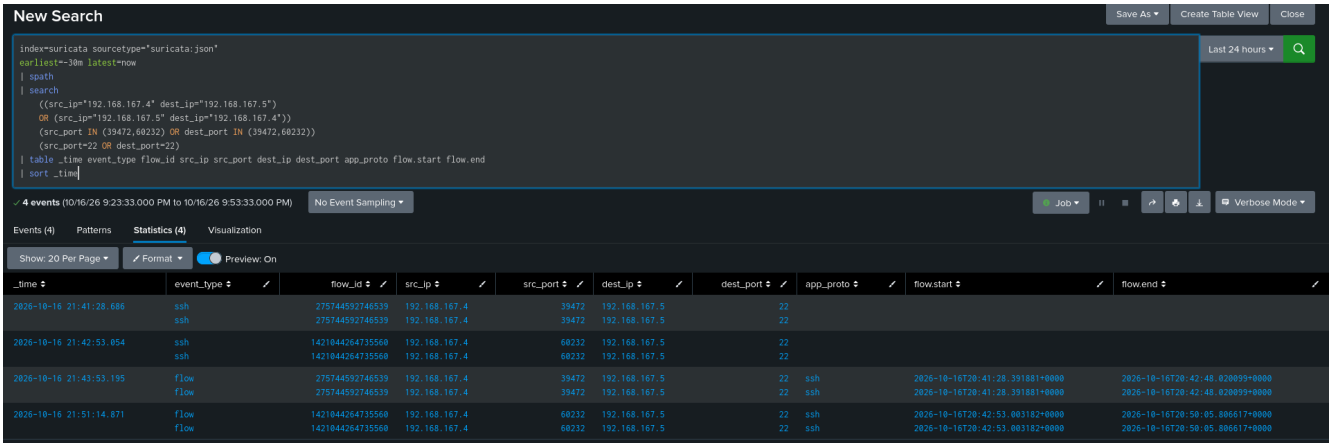


Figure 6 Suricata SSH and flow evidence for the same source and destination E07

Assessment

The cross-source investigation is complete for the supplied test. Host and network records agree on the client and server, distinguish the two connections by source port, and provide compatible timestamps. No Suricata attack signature is required for this correlation because SSH and flow events provide connection evidence.

Framework context

Repeated failed passwords are behaviour relevant to MITRE ATT&CK T1110.001 Password Guessing. The Windows detection is a lab simulation of that behaviour, not attribution of an adversary. A single failed SSH password followed by an accepted password is insufficient to label the SSH test brute force. [R4]

Central collection and event analysis are consistent with the intent of NIST CSF Detect outcomes DE.CM and DE.AE. This is an illustrative mapping only; the lab does not demonstrate organisation-wide implementation or certification. [R5]

Detection searches and reproduction notes

Windows five failures in ten minutes

```
index=windows sourcetype="XmlWinEventLog:Security"
"<EventID>4625</EventID>" earliest=-10m latest=now
| rex field=_raw "<Data Name='TargetUserName'>(?<user>[^<]+)</Data>"
| rex field=_raw "<Data Name='IpAddress'>(?<src_ip>[^<]+)</Data>"
| stats count as failures by host user src_ip
| where failures >= 5
```

Schedule: * * * * *; dispatch earliest: -10m; latest: now. Trigger when the number of result rows is greater than zero. Add to Triggered Alerts. The threshold of five is applied inside SPL, so the UI result threshold remains zero. Single-quoted XML extraction matches the supplied events; managed field extraction should be validated before wider deployment. [R2]

Ubuntu authentication records

```
index=linux source="/var/log/auth.log" "192.168.167.4"
("Failed password" OR "Accepted password")
| table _time host _raw
| sort 0 _time
```

For reproduction, select 16 October 2026, 20:40–20:53 UTC, or 21:40–21:53 BST if using the same display timezone as the supplied screenshots. Search time interpretation depends on the Splunk user timezone.

Suricata connections for the same hosts

```
index=suricata sourcetype="suricata:json"
| spath
| search event_type IN ("ssh", "flow")
| where (src_ip="192.168.167.4" AND dest_ip="192.168.167.5")
    OR (src_ip="192.168.167.5" AND dest_ip="192.168.167.4")
| where src_port=22 OR dest_port=22
| table _time event_type flow_id src_ip src_port dest_ip dest_port
    flow.start flow.end
| sort 0 _time
```

Use the same absolute time window. This search is provided for reproduction; the completed finding relies on the supplied results. If JSON fields already extract correctly, validate the extraction before adding spath to avoid duplicate field values.

Inventory and alert diagnostics

```
(index=windows OR index=linux OR index=suricata OR index=_internal)
earliest=-24h latest=now
| stats count latest(_time) as last_seen by index host source sourcetype
| convert ctime(last_seen)
```

If an alert is absent, compare the saved search dispatch bounds, scheduler execution, result count and configured action. A manual search returning results alone does not prove that a scheduled alert executed.

Recommendations and evidence register

Priority	Recommended next action	Reason
1	Retain the final inputs, outputs and saved-search settings with the report.	Make the validated setup reproducible.
1	Apply tested throttling by host, user and source IP.	Reduce repeated notifications from overlapping windows.
1	Validate NTP and record the Splunk user timezone.	Keep future cross-source comparisons consistent.
2	Test visibility for every required VM traffic path.	Establish the actual scope of sensor coverage.
2	Review TLS, access roles, retention and licence volume.	These production controls were not demonstrated.
2	Validate standard XML and JSON field extractions.	Reduce brittle regex and multivalue display issues.

Additional process telemetry, automated containment, alert delivery to email or a case platform, continuous packet-loss performance and production coverage were not established by the supplied evidence. The report makes no measured detection-time improvement or compliance claim.

ID	Evidence item	Result or record
E01	Windows search and supplied raw 4625 XML	6 failures; record 24120; NewUser loopback test
E02	Triggered Alerts screenshot 19:01	3 scheduled entries after dispatch correction
E03	Kestrel SOC dashboard screenshot 19:31	Windows authentication panels and activity chart
E04	Supplied Windows System event XML 13 October 2026	Service Control Manager event 7040 for BITS
E05	Ubuntu collection screenshot 19:25	auth.log 165; syslog 5,614 in that window
E06	Host SSH screenshot 21:52 and command output	Failed password then accepted password
E07	Network SSH screenshot 21:53	2 SSH events and 2 corresponding flow events
E08	ICMP test screenshot	3 echo requests and 3 lab alerts
E09	Suricata dashboard and supplied alert table	3 ICMP alerts and 2 DHCP hostname alerts
E10	Platform inventory screenshot 19:14	63,692 events; 42 grouped combinations

Evidence was supplied as screenshots and pasted output. Original raw exports and cryptographic hashes were not supplied. This report is a portfolio validation record rather than a forensic chain-of-custody package. Companion deliverables are the final log source inventory workbook and the project presentation.

Technical references

The primary evidence is the lab output listed in the evidence register. The following vendor and framework references support interpretation and configuration context.

R1 Microsoft Windows event 4625

Failed logon fields, logon types and status interpretation.

https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/event-4625

R2 Splunk search endpoint descriptions

Saved searches and scheduling configuration context.

https://help.splunk.com/en/splunk-enterprise/rest-api-reference/10.2/search-endpoints/search-endpoint-descriptions

R3 Suricata EVE JSON format

SSH, flow and alert record structures. The linked documentation is version 8.0.2; the observed sensor reported 8.0.7.

https://docs.suricata.io/en/suricata-8.0.2/output/eve/eve-json-format.html

R4 MITRE ATT&CK Password Guessing

Technique T1110.001. Mapping describes behaviour relevant to the lab detection.

https://attack.mitre.org/techniques/T1110/001/

R5 NIST Cybersecurity Framework 2.0 overview

Framework context for monitoring and event analysis.

https://www.nist.gov/system/files/documents/2024/03/20/March20_2024_NISTCSF2.0_SMBQSG.Overview.pdf

R6 Ubuntu OpenSSH server

Server configuration and administration reference.

https://ubuntu.com/server/docs/openssh-server/

Report conclusion

The lab demonstrates endpoint and network log ingestion, a functioning Windows failed logon alert, a usable SOC dashboard and a documented SSH investigation across two sources. The evidence supports completion of the core home-lab monitoring exercise, with the recommendations above defining the next steps toward a more robust deployment.